

Proposal: Cluster-Based Anomaly Detection for Aqua Blend

K-Means++ Water-Quality Profiling with Data-Driven Distance Threshold Optimisation

Aqua Blend — Technical Design Proposal

Abstract

This proposal presents a clustering-based anomaly detection component for Aqua Blend. The proposed method does not operate on the user-defined optimisation scenario itself. Instead, it uses the underlying historical water-quality database, where variables such as pH, turbidity, total dissolved solids (TDS), conductivity, hardness and other source-water measurements form a multivariate feature vector. K-Means++ is used to learn recurring normal water-quality profiles, and a new observation is assessed by measuring its distance to the nearest learned centroid.

The main contribution is a threshold-learning strategy for anomaly detection. Rather than adopting an arbitrary fixed rule such as the 95th percentile, the proposal learns a separate distance threshold A_k for each cluster. Candidate thresholds are evaluated on a validation dataset containing normal records and labelled synthetic anomalies. The selected thresholds are then optimised using precision, recall and F_β , allowing Aqua Blend to explicitly control the trade-off between false alarms and missed anomalies. The resulting method is computationally light, explainable and suitable for integration into Aqua Blend's existing analysis pipeline.

1 Introduction and Problem Context

Aqua Blend already uses structured system data to support optimisation. The purpose of the proposed anomaly detector is different from the Mixed-Integer Linear Programming (MILP) optimisation model. The detector answers the following question: *does a newly observed water-quality record resemble the normal multivariate patterns represented in the historical database, or is it unusual enough to warrant further investigation?*

In anomaly detection, an anomaly is an observation that differs meaningfully from the expected pattern of the data [3]. In the Aqua Blend setting, an anomalous record may correspond to a genuine change in source-water conditions, a rare but plausible operating state, a sensor error, a data-entry error, or another observation that is poorly represented in the historical dataset. Consequently, the anomaly detector should be treated as a decision-support and data-quality mechanism rather than as proof of a physical failure. Clustering is appropriate because the water-quality database may contain more than one legitimate normal operating profile. For example, water drawn from different sources or measured under different seasonal or treatment conditions may naturally occupy different regions in feature space. A single global mean would therefore be too simplistic. Cluster-based outlier detection is well established in the literature, including methods that use the relationship between cluster structure and local deviation to identify unusual observations [4]. The proposed Aqua Blend design adopts the same broad intuition while keeping the resulting anomaly score transparent and easy to interpret.

1.1 Scope of the proposed component

The proposed component takes the historical database as input rather than the optimisation scenario object passed to the MILP solver. For observation i , define the raw feature vector as

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{ip}]^T \in \mathbb{R}^p, \quad (1.1)$$

where p is the number of selected numerical water-quality variables. A representative feature vector is

$$\mathbf{x}_i = [\text{pH}, \text{turbidity}, \text{TDS}, \text{conductivity}, \text{hardness}, \dots]^T. \quad (1.2)$$

The exact feature list should be chosen from fields that are consistently available, numerically meaningful and sufficiently reliable in the Aqua Blend database. The detector should return, at minimum, the nearest learned cluster, the distance from the new observation to that cluster centroid, the relevant threshold A_k , an anomaly margin or normalised anomaly score, a final label (NORMAL or ANOMALY), and a ranked explanation of which features contributed most strongly to the distance.

2 Data Preparation and Mathematical Representation

2.1 Feature scaling

K-Means minimises Euclidean distance, so variables with larger raw numerical ranges can dominate the clustering process. For example, conductivity or TDS may numerically outweigh pH unless all variables are first placed onto a comparable scale. The model should therefore be trained in a standardised feature space.

For feature j , let μ_j and σ_j denote the training-set mean and standard deviation respectively. Each raw value is transformed using

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}. \quad (2.1)$$

The resulting standardised observation is

$$\mathbf{z}_i = [z_{i1}, z_{i2}, \dots, z_{ip}]^\top. \quad (2.2)$$

It is important that the scaling parameters in Equation (2.1) are calculated using the training partition only. Validation and test observations must be transformed using the stored training values of μ_j and σ_j . Recomputing the scaler with later data would leak information into the model-development process.

2.2 Dataset partitions

The overall dataset should be separated into three logical partitions:

1. **Training-normal set:** historical observations believed to represent normal water-quality behaviour. This set is used to fit the scaler and the K-Means++ centroids.
2. **Validation set:** held-out normal observations combined with labelled synthetic anomalies. This set is used to learn the anomaly thresholds.
3. **Test set:** data reserved for final evaluation. If real labelled anomalies are available, they should be prioritised here. Otherwise, reported performance must be described as performance against the synthetic anomaly design rather than direct evidence of real-world anomaly-detection effectiveness.

This separation is essential because the anomaly threshold itself is a learned decision parameter. Using the same records for both threshold learning and final reporting would produce an overly optimistic result.

2.3 Notation summary

Notation	Domain	Description
$\mathbf{x}_i$	$\mathbb{R}^p$	Raw feature vector for observation i
$\mathbf{z}_i$	$\mathbb{R}^p$	Standardised feature vector for observation i
K	$\mathbb{N}$	Number of clusters
$\mathbf{c}_k$	$\mathbb{R}^p$	Centroid of cluster k
$g(i)$	$\{1, \dots, K\}$	Cluster assignment of training observation i
$d_k(\mathbf{z})$	$\mathbb{R}_{\geq 0}$	Distance from observation $\mathbf{z}$ to centroid $\mathbf{c}_k$
k^*	$\{1, \dots, K\}$	Index of nearest centroid for a new observation
A_k	$\mathbb{R}_{\geq 0}$	Learned anomaly threshold for cluster k
$M(\mathbf{z})$	$\mathbb{R}$	Signed anomaly margin for a new observation
$S(\mathbf{z})$	$\mathbb{R}_{\geq 0}$	Normalised anomaly score for a new observation
y_i	$\{0, 1\}$	Ground-truth label, where 0 = normal and 1 = anomaly

Table 1: Notation used throughout the proposal.

3 Learning Normal Profiles with K-Means++

3.1 Clustering objective

Assume the normal training set contains n standardised observations and that K clusters are fitted. Let cluster k have centroid $\mathbf{c}_k \in \mathbb{R}^p$. K-Means seeks the partition that minimises the within-cluster sum of squared distances:

$$J = \sum_{i=1}^n \|\mathbf{z}_i - \mathbf{c}_{g(i)}\|_2^2, \quad (3.1)$$

where $g(i) \in \{1, \dots, K\}$ denotes the cluster assigned to training observation i . Equivalently, the assignment rule is

$$g(i) = \arg \min_{k \in \{1, \dots, K\}} \|\mathbf{z}_i - \mathbf{c}_k\|_2. \quad (3.2)$$

K-Means can be sensitive to its initial centroid positions. K-Means++ improves the initialisation by selecting centres according to a distance-aware probability distribution [1]. If $D(\mathbf{z})$ is the distance from point $\mathbf{z}$ to its nearest already selected centre, then a new centre is sampled with probability proportional to

$$P(\mathbf{z}) = \frac{D(\mathbf{z})^2}{\sum_{\mathbf{z}' \in X} D(\mathbf{z}')^2}. \quad (3.3)$$

For Aqua Blend, K-Means++ is therefore preferred over purely random initialisation.

3.2 Selecting the number of clusters

The value of K should be chosen empirically rather than fixed arbitrarily. A practical search range is $K = 2, \dots, 10$, although the final range can be adjusted to match the size and diversity of the database. One useful diagnostic is the silhouette coefficient introduced by Rousseeuw [5]:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad (3.4)$$

where $a(i)$ is the average distance from observation i to observations in its own cluster, and $b(i)$ is the smallest average distance from observation i to observations in another cluster. Values near 1 indicate strong separation, values near 0 indicate boundary observations, and negative values suggest that another cluster may provide a better fit.

Silhouette score alone should not determine the final choice of K . Cluster stability across different random seeds, minimum cluster size, and domain interpretability should also be considered. The final aim is to learn meaningful recurring water-quality profiles rather than simply maximising a single metric.

4 Distance-Based Anomaly Detection Layer

4.1 Distances from a new sample to every centroid

After training, consider a new standardised water-quality observation $\mathbf{z}$. Its Euclidean distance to each centroid is

$$d_k(\mathbf{z}) = \|\mathbf{z} - \mathbf{c}_k\|_2 = \sqrt{\sum_{j=1}^p (z_j - c_{kj})^2}, \quad k = 1, \dots, K. \quad (4.1)$$

If $K = 3$, the detector will produce the distance vector

$$[d_1, d_2, d_3]. \quad (4.2)$$

Suppose a new observation gives

$$[d_1, d_2, d_3] = [0.42, 1.10, 1.85]. \quad (4.3)$$

The nearest cluster is identified by

$$k^* = \arg \min_k d_k = 1, \quad (4.4)$$

and the relevant nearest-centroid distance is

$$d_{\min} = d_{k^*} = 0.42. \quad (4.5)$$

K-Means alone would assign the observation to Cluster 1 on the basis of Equation (4.4). However, clustering by itself does not determine whether 0.42 is a typical distance for Cluster 1 or an unusually large one. This motivates a separate anomaly-decision layer.

4.2 Cluster-specific thresholds

Let A_k denote the anomaly threshold associated with cluster k . The proposed binary anomaly rule is

$$\hat{y}(\mathbf{z}) = \begin{cases} 1, & d_{k^*}(\mathbf{z}) > A_{k^*}, \\ 0, & d_{k^*}(\mathbf{z}) \leq A_{k^*}, \end{cases} \quad (4.6)$$

where 1 corresponds to an anomaly and 0 corresponds to a normal observation.

A separate threshold is learned for each cluster because different clusters may have different spreads. A compact cluster should have a tighter threshold than a naturally more variable cluster. This is one of the main reasons a single global cutoff is not recommended.

The signed anomaly margin is defined as

$$M(\mathbf{z}) = d_{k^*}(\mathbf{z}) - A_{k^*}. \quad (4.7)$$

If $M(\mathbf{z}) \leq 0$, the point lies within the accepted region of its nearest cluster. If $M(\mathbf{z}) > 0$, the point lies outside that learned boundary.

A normalised cluster-independent score is also useful:

$$S(\mathbf{z}) = \frac{d_{k^*}(\mathbf{z})}{A_{k^*}}. \quad (4.8)$$

Under Equation (4.8), $S < 1$ indicates that the point lies inside the decision boundary, $S = 1$ lies exactly on the boundary, and $S > 1$ indicates an anomaly.

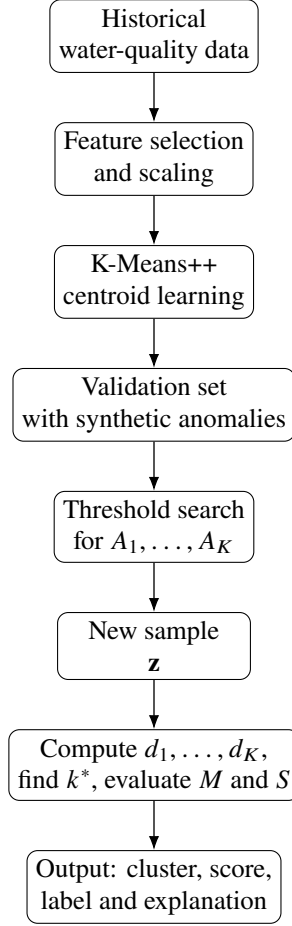


Figure 1: Conceptual workflow of the proposed clustering-based anomaly detection pipeline for Aqua Blend.

5 Synthetic Anomaly Generation and Threshold Optimisation

5.1 Why a fixed percentile is not enough

A percentile threshold such as the 95th percentile can be used as an unsupervised baseline. For example, one could set A_k to the 95th percentile of the training distances for points assigned to cluster k . However, the number 95 is not inherently optimal for Aqua Blend. A high percentile may reduce false alarms but miss near-boundary anomalies. A lower percentile may capture more anomalies but produce an impractical number of false positives.

For this reason, the present proposal treats a percentile-based threshold only as a baseline. The preferred method is to learn A_k using a validation dataset with explicit labels.

5.2 Synthetic anomaly generation

Real anomaly labels are often scarce in practical anomaly-detection problems [3]. Aqua Blend can therefore begin with a controlled synthetic validation dataset. The synthetic records should not be arbitrary random values. Instead, they should be generated from the statistical ranges and relationships observed in the historical dataset, then perturbed in a controlled manner. Each validation record receives a ground-truth label

$$y_i = \begin{cases} 0, & \text{normal,} \\ 1, & \text{synthetic anomaly.} \end{cases} \quad (5.1)$$

The following anomaly categories are proposed:

1. **Single-feature excursion:** one variable is pushed beyond its normal range while others remain plausible.
2. **Multivariate edge anomaly:** several variables are shifted moderately so that no single value is extreme, but the combined point lies close to or just outside the learned cluster boundary.
3. **Combination anomaly:** individual values remain within broad historical ranges, but the joint combination is rare or poorly represented in the training data.
4. **Strong outlier:** one or more variables are shifted substantially to create clear deviations that should be easy to detect.

A practical CSV structure for the validation dataset is shown in Table 2.

ID	pH	Turbidity	TDS	Conduct.	Anomaly type	Label
N001	7.10	1.4	280	420	normal	0
N002	6.85	2.0	315	465	normal	0
A001	7.05	7.8	300	450	feature excursion	1
A002	6.25	4.6	510	760	multivariate edge	1

Table 2: Illustrative schema for the validation dataset containing normal observations and labelled synthetic anomalies. The values are examples only.

5.3 Learning the optimal threshold

For each validation sample i , first determine its nearest cluster

$$k_i^* = \arg \min_k d_{ik}, \quad (5.2)$$

and its nearest-centroid distance

$$r_i = d_{i,k_i^*}. \quad (5.3)$$

For cluster k , define a set of candidate thresholds

$$\mathcal{A}_k = \{A_k^{(1)}, A_k^{(2)}, \dots, A_k^{(m)}\}. \quad (5.4)$$

Each candidate may be obtained from a fine numerical grid or from the sorted unique validation distances associated with the cluster.

The earlier design idea of computing terms such as $d - A$ is retained in a more precise form through the anomaly margin in Equation (4.7). However, the threshold becomes meaningful only when evaluated against labelled data. A second machine-learning model is therefore unnecessary; the threshold itself can be learned directly by searching over $\mathcal{A}_k$.

For each candidate threshold, predictions are compared with the labels in Equation (5.1), producing the confusion-matrix counts: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Precision is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (5.5)$$

recall is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (5.6)$$

and the F_1 score is

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (5.7)$$

If Aqua Blend wishes to prioritise anomaly recall more strongly than precision, the more general F_β score is

$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \times \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}}, \quad \beta > 1. \quad (5.8)$$

A cluster-specific optimal threshold may then be selected using

$$A_k^* = \arg \max_{A \in \mathcal{A}_k} F_\beta(A). \quad (5.9)$$

An alternative engineering rule is to impose a minimum required recall $R_{\min}$ and then choose the threshold with the smallest false-positive rate subject to

$$\text{Recall}(A) \geq R_{\min}. \quad (5.10)$$

This makes the operational trade-off explicit.

5.4 Worked example

Assume that the learned thresholds are

$$[A_1^*, A_2^*, A_3^*] = [0.68, 0.83, 0.57]. \quad (5.11)$$

A new observation produces

$$[d_1, d_2, d_3] = [0.73, 1.28, 1.91]. \quad (5.12)$$

The nearest cluster is Cluster 1. Therefore, the anomaly margin is

$$M = 0.73 - 0.68 = 0.05, \quad (5.13)$$

and the normalised anomaly score is

$$S = \frac{0.73}{0.68} \approx 1.074. \quad (5.14)$$

Since both $M > 0$ and $S > 1$, the record is labelled as **ANOMALY**. The relatively small positive margin in Equation (5.13) also indicates that this is a near-boundary anomaly rather than an extreme outlier.

6 Interpretability, Integration and Evaluation

6.1 Feature-level explanation

The total squared distance to the selected centroid can be decomposed into feature-level contributions:

$$d_{k^*}^2 = \sum_{j=1}^p (z_j - c_{k^*j})^2. \quad (6.1)$$

A normalised contribution score for feature j is then

$$q_j = \frac{(z_j - c_{k^*j})^2}{\sum_{\ell=1}^p (z_\ell - c_{k^*\ell})^2}. \quad (6.2)$$

The values satisfy approximately

$$\sum_{j=1}^p q_j = 1. \quad (6.3)$$

Ranking the q_j values allows Aqua Blend to explain which variables contributed most to the anomaly score. This should be interpreted as a mathematical contribution to centroid distance rather than a claim of physical causation.

6.2 Runtime integration

The proposed runtime workflow is summarised below:

1. Load validated historical water-quality records.
2. Select the agreed numerical features and apply the stored scaler.
3. Use the stored K-Means++ model to compute $d_1, \dots, d_K$.
4. Identify the nearest cluster k^* using Equation (4.4).
5. Evaluate the anomaly margin M and normalised score S using Equations (4.7) and (4.8).
6. Return the cluster index, threshold, score, binary label and feature-level contribution summary.

A possible machine-readable response is shown in ?? 1.

Listing 1: Illustrative JSON response returned by the anomaly detector.

```
{
  "nearest_cluster": 1,
  "distance": 0.73,
  "threshold": 0.68,
  "anomaly_margin": 0.05,
  "anomaly_score": 1.074,
  "label": "ANOMALY",
  "top_distance_features": ["turbidity", "conductivity"]
}
```

6.3 Evaluation plan

Overall accuracy alone should not be the primary evaluation metric because anomaly datasets are often imbalanced [6]. The evaluation should report, at minimum:

- precision, recall and F_1 or the chosen F_β ;
- the confusion matrix;
- performance by synthetic anomaly type and anomaly severity;
- false-positive rate on held-out normal observations;
- the distribution of anomaly margins M ;
- cluster-level results to determine whether any threshold is weak or unstable;
- sensitivity to the choice of K and to multiple K-Means++ initialisations.

The optimised detector should also be compared against a simple percentile-based baseline. This demonstrates whether the labelled threshold search materially improves performance rather than merely assuming that it does.

6.4 Limitations and future work

Several limitations should be acknowledged. First, K-Means represents clusters using Euclidean geometry and is therefore most suitable when normal data occupy reasonably compact regions in standardised feature space. If the data exhibit strong density variation or highly irregular shapes, other anomaly-detection methods may be more appropriate. Local Outlier Factor (LOF), for example, was developed specifically to compare local densities and detect observations that are sparse relative to their neighbours [2].

Second, synthetic anomalies are useful for controlled development but do not guarantee performance on every real anomaly pattern. As Aqua Blend accumulates operational evidence, real labelled anomalies should be incorporated into validation and test data. Third, the learned centroids and thresholds may become stale if the distribution of the water-quality database drifts over time. The detector should therefore be versioned and periodically re-evaluated rather than retrained automatically without review.

Finally, an anomaly flag means that an observation differs from the learned historical patterns. It does not automatically imply unsafe water, a process failure, or a regulatory breach. Those interpretations remain separate downstream tasks.

7 Conclusion

This proposal presents a mathematically transparent anomaly detection layer for Aqua Blend based on clustering of historical water-quality records. K-Means++ is used to learn recurring normal water-quality profiles from standardised feature vectors. A new observation is then compared with the learned centroids, and its nearest-centroid distance becomes the basis of anomaly scoring.

The central design choice is the use of cluster-specific learned thresholds A_k^* . Rather than fixing a global value or relying on an arbitrary percentile such as 95%, the proposal optimises these thresholds against labelled validation data containing both normal records and controlled synthetic anomalies. This makes the false-positive versus false-negative trade-off measurable and tunable for the needs of Aqua Blend.

The final decision process is compact and interpretable:

$$\mathbf{x} \rightarrow \mathbf{z} \rightarrow [d_1, \dots, d_K] \rightarrow k^* \rightarrow d_{k^*} - A_{k^*}^* \rightarrow \{\text{NORMAL}, \text{ANOMALY}\}.$$

Because the method can also return the relevant threshold, anomaly margin, normalised score and feature-level distance contributions, it integrates naturally into Aqua Blend’s broader optimisation and AI-assisted interpretation workflow. As future work, the same framework can be compared with density-based or one-class anomaly-detection methods once sufficient operational evidence becomes available.

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